# Neural Bellman Operators
# Author Response to the Advisory Referee

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## Response to the controlling referee advisory

This is an author revision of Neural Bellman Operators for the Econometrica numerical-methods objective. The controlling owner-commissioned advisory report is R97 at `9723c6cd2cc8b3e071422f3e522d4a843946db63`. It is an advisory on the named source, not an Econometric Society editorial decision. It reviewed manuscript `f802fe64d91d9f5c1c8f86123b0fbe114e622279`, not the subsequent author revisions. The current source baseline is the complete R125 author package at `eda26b02fbfb1e02807bc0cee1c0015d62d59e56`. The actual published R126 scientific boundary is `769e3eb3ac08beaa7bb6778c1676f15e1f8d6d2b`, with immutable readback before new economics. The original Bellman objective, continuation critics, feasible actors and economic research identity remain central. Each response separates the inherited amendment, current author action, actual evidence status and remaining scientific requirement. No source audit, smaller fitting residual or unchanged conventional winner is treated as completion of the research objective.

**Response map status.** Both scientific episodes, independent fitting, exact support checking and the fresh offline source gate have executed. Final native documents are identified by the package build/hash receipts; remote CI is reported only through an actual run receipt.

All 35 report lineages and 684 overlapping historical extraction records remain source-bound in the retained development companion; no blanket closure is claimed.

All new statements below refer to the actual F126 episode and its separately identified audits. The deterministic census is not a population sample. The frozen independent audit leaves 307 tighter reported-regret-bound confirmations and 1,113 containments of the independent policy interval inside the reported interval unresolved, despite zero contradictions. Its qualification-named fields test regret only. A separate saved-bound width audit now verifies all 1,584 ordinary and 1,175 stronger complete qualification gates without new economics; that audit does not resolve the other open comparisons. The exact full-support solver produces no additional qualified gain. Original adverse timing, actor bytes, negative adoption conditions and historical failures remain retained. Direct versus the strong warm-adaptive control has 24 strict gains, 26 losses, 257 exact ties and 215 unresolved pairs, with no resolved mean-cost ordering. The distinct C algorithm preserves every complete scientific record while reducing certificate computation costs; it remains more costly than equally cached warm adaptive in the pooled nine-state comparison.

**Separately identified follow-up episode.** R126C exact-geometry caching episode (source `9c6246ecad31acd6a937a2d8ad4710bcabc8c801`): Public source and readback preceded all C economics; the paired primary, serial timing and full scientific-equality verification completed successfully.

The separately source-frozen R126C paired episode executed 3,168 new primary and 1,056 new timed decisions. All 4,224 complete scientific records agree with their original reference after only the prospectively declared clock and run-metadata exclusions. The same within-decision geometry cache is available to every neural and conventional service and is reset before each next paid decision. Complete parent-allocated nine-state costs change from 16.2466 to 15.9305 seconds for fused control, 35.1063 to 27.4172 for warm adaptive, 68.8494 to 33.5117 for direct terminal, and 7.7973 to 7.7847 for grid DP. Direct cost falls by 51.326 percent but remains 22.229 percent above equally cached warm adaptive. All first-campaign adverse costs remain unchanged. These are descriptive costs, not timing confidence intervals. The C episode supplies 192 sufficient neural adoption regions, all empty, with 192 exact same-policy conventional cost blockers; all 192 optimistic outer regions remain nonempty. Exact reuse improves common certificate computation without establishing a neural-specific representation, information or policy advantage. The C protocol was designed after observing the first adverse result and frozen before its own executions; the first result and all original actors, data and policies remain preserved.

### B1. Neural specificity remains unestablished

**Original request.** Validity, recursive qualification, the AVaR enclosure, the information lower bound, and the fallback are representation-neutral. The ReLU family is not the complete-work winner and is never certified adopted over quadratic least squares.

**Inherited R125 amendment.** We retain the original NBO continuation critic, feasible actor, recursive Bellman target and economic decision problem. Theorem fullinterpolation125 gives an exact finite convex representation of the information in every paid original quadratic support, including an uncertain but globally valid supporting slope. This closes the explicit raw-support omission in the R124 information class. Theorem paidwork125 links original support precision to feasible action localization, its declared work bound and a separate returned-policy gate. Both mechanisms are available to every representation under the same information. The R125 protocol retains all 56 original actors and the exactly matched ordinary-hinge control.

**Current R126 author action.** The revision retains the original Neural Bellman Operator, continuation critic, feasible actor and original economic objective. The finite full-support rational-cut algorithm uses all paid quadratic supports to sharpen an original-action decision calculation; the proposed stopping bounds concern original regret. These operations are available to the neural and equally informed conventional services. The new census retains all 56 historical actors and their original information/feature identities.

**Evidence status.** Executed first-campaign evidence and proved components; the stated scientific limits remain open.

The main and supplement now state and prove finite rational-cut validity, residual correction over the declared box, the original-action regret enclosure and its finite-grid accuracy, and the direct-service projected gap, constrained optimizer bracket, separate regret/value-width gates and strict same-objective policy replacement. General convexity, interpolation and weak-duality ingredients are identified as inherited or classical.

The frozen paid-support campaign executed all 74 pools, 888 contrasts and 5,920 LP calls. Exact independent reconstruction checked 33,331 cuts and 148 candidate regret records. It found 48 strictly sharper contrasts and six sharper regret bounds, but zero newly qualified policy gains and

zero changed primary policies. No new economic query was made, and no finite solver optimality was certified. The six improved regret bounds concern capacity actions that remain unqualified, so they are not a delivered economic gain.

The executed first campaign retains all 56 acquired actors, both online risk controls and the action-grid baseline across 1,584 original-model decisions. All 96 sufficient neural adoption regions are empty. All 96 optimistic outer regions remain nonempty, while every case has an exact same-policy conventional cost blocker under its recorded acquisition and timing ranges. This supplies no neural-specific dominance claim.

The C episode supplies 192 sufficient neural adoption regions, all empty, with 192 exact same-policy conventional cost blockers; all 192 optimistic outer regions remain nonempty. Exact reuse improves common certificate computation without establishing a neural-specific representation, information or policy advantage.

**Remaining scientific requirement.** Neural-specific superiority requires a strict benefit over equally informed conventional representations after acquisition, certification and complete-service costs. A common safeguard or a successful package is insufficient.

**Source routes.** `new126/certified_support_main126.tex`, `new126/certified_support_proofs126.tex`, `new126/decision_stopping_main126.tex`, `new126/decision_stopping_proofs126.tex`, `results126/support01/SUMMARY126.json`, `results126/support-audit01/SUPPORT_VERIFICATION126.json`, `new126/support_results126.tex`, `results126/report01/RESULTS126.md`, `results126/report01/SUMMARY126.json`, `results126/primary01/EXECUTION126.json`, `results126/independent01/SUMMARY126.json`, `SOURCE_FREEZE126C.json`, `BOUNDARIES126C.json`, `new126/cached_envelopes_main126c.tex`, `new126/cached_envelopes_proofs126c.tex`, `results126c/verification01/EXECUTION126.json`, `results126c/report01/SUMMARY126C.json`, `results126c/report01/RESULTS126C.md`.

### B2. The information-complexity result is too narrow for the platform claim

**Original request.** The theorem is a specialized deterministic one-dimensional oracle result. It does not establish randomized, multivariate, nonuniform-law, or end-to-end Bellman complexity lower bounds.

**Inherited R125 amendment.** Theorem fullinterpolation125 identifies the exact worst compatible action contrast for a fixed pair under a scalar, smooth strongly convex, complete-support information interface. It explicitly excludes additional economic formulas unless encoded. The support inequalities are convex bounded-domain conjugate constraints; necessity, constructive sufficiency, compactness, degenerate cases and rational residual certification are proved. The terminal bracket proposition separately gives a matching randomized-action minimax result for its reduced stationary-bracket interface.

**Current R126 author action.** The new finite-support calculation states its actual information class, finite convex representation and exact checking interface. The regret-stopping analysis connects the declared information to the original decision. We keep these results distinct from the inherited deterministic univariate point-value lower bound and identify the classical interpolation and convex-duality ingredients.

**Evidence status.** Executed or proved components with explicit remaining scientific scope.

The main and supplement now state and prove finite rational-cut validity, residual correction over the declared box, the original-action regret enclosure and its finite-grid accuracy, and the direct-service projected gap, constrained optimizer bracket, separate regret/value-width gates and strict same-objective policy replacement. General convexity, interpolation and weak-duality ingredients are identified as inherited or classical.

The frozen paid-support campaign executed all 74 pools, 888 contrasts and 5,920 LP calls. Exact independent reconstruction checked 33,331 cuts and 148 candidate regret records. It found 48 strictly sharper contrasts and six sharper regret bounds, but zero newly qualified policy gains and zero changed primary policies. No new economic query was made, and no finite solver optimality was certified. The six improved regret bounds concern capacity actions that remain unqualified, so they are not a delivered economic gain.

The geometry-reuse theorem supplies a finite O(s^3) rational-operation construction account and exact within-service reuse bounds, with key formation, requested clipped output, table operations and rational bit costs explicitly separated. It proves scientific identity, not a uniform speed advantage, a new information lower bound, or a high-dimensional continuous-risk complexity theorem. The paired C campaign measures the actual complete costs.

**Remaining scientific requirement.** A general randomized, multivariate, nonuniform-law, end-to-end Bellman complexity theorem remains separate. Finite paid-support optimality cannot be relabeled as such a theorem.

**Source routes.** `new126/certified_support_main126.tex`, `new126/certified_support_proofs126.tex`, `new126/decision_stopping_main126.tex`, `new126/decision_stopping_proofs126.tex`, `results126/support01/SUMMARY126.json`, `results126/support-audit01/SUPPORT_VERIFICATION126.json`, `new126/support_results126.tex`, `SOURCE_FREEZE126C.json`, `BOUNDARIES126C.json`, `new126/cached_envelopes_main126c.tex`, `new126/cached_envelopes_proofs126c.tex`, `results126c/verification01/EXECUTION126.json`, `results126c/report01/SUMMARY126C.json`, `results126c/report01/RESULTS126C.md`.

### B3. The representation result is not decision-theoretic

**Original request.** The ridge example separates restricted approximation classes but does not separate policies, service work, or economic outcomes. A conventional method with the same hinge is equivalent.

**Inherited R125 amendment.** The new optimized quantity is the original action contrast q(b)-q(a), followed by the explicit implemented-future allowance and the original regret and objective-width gates. It is therefore a decision quantity. The exact witness changes the worst compatible contrast from -1/4 to -5/8 when a previously discarded support is retained, and states exactly which conventional calculation already achieves the same result. No new neural representation separation is claimed.

**Current R126 author action.** The action-level certificate is evaluated at feasible decisions in the original economic problem. ReLU and the ordinary-hinge arm retain the same fixed features, paid labels and stored readout, so their geometric equivalence is explicit. The new decision solver cannot create neural specificity merely by improving both encodings.

**Evidence status.** Executed first-campaign evidence and proved components; the stated scientific limits remain open.

The main and supplement now state and prove finite rational-cut validity, residual correction over the declared box, the original-action regret enclosure and its finite-grid accuracy, and the direct-service projected gap, constrained optimizer bracket, separate regret/value-width gates and strict same-objective policy replacement. General convexity, interpolation and weak-duality ingredients are identified as inherited or classical.

The frozen paid-support campaign executed all 74 pools, 888 contrasts and 5,920 LP calls. Exact independent reconstruction checked 33,331 cuts and 148 candidate regret records. It found 48 strictly sharper contrasts and six sharper regret bounds, but zero newly qualified policy gains and zero changed primary policies. No new economic query was made, and no finite solver optimality was certified. The six improved regret bounds concern capacity actions that remain unqualified, so they are not a delivered economic gain.

All seven representation families retain the same source-bound actor and information identities. The matched ReLU/ordinary-hinge comparison remains an exact-geometry control. Independent comparisons of the direct service against its two completed controls yielded 238 strict gains, 39 strict losses, 517 exact complete-policy ties and 250 unresolved orderings across 1,044 pairs. These compare services within the original economic criterion, not architectural superiority.

The strong warm-adaptive comparator must be separated from fused control. Among 522 direct-versus-warm pairs, the exact independent audit finds 24 strict gains, 26 strict losses, 257 exact policy ties and 215 unresolved orderings; its deterministic equal-weight mean difference is enclosed by [-7.002e-6, 7.125e-6], which does not resolve a mean-cost ordering. Against fused control, 522 pairs yield 214 gains, 13 losses, 260 ties and 35 unresolved orderings, with mean difference enclosed by [-3.681e-4, -3.530e-4]. These are bounds for the fixed census, not population confidence intervals.

**Remaining scientific requirement.** No action-regret or complete-work separation is inferred from an approximation example that an equally informed ordinary hinge also realizes.

**Source routes.** `new126/certified_support_main126.tex`, `new126/certified_support_proofs126.tex`, `new126/decision_stopping_main126.tex`, `new126/decision_stopping_proofs126.tex`, `results126/support01/SUMMARY126.json`, `results126/support-audit01/SUPPORT_VERIFICATION126.json`, `new126/support_results126.tex`, `results126/report01/RESULTS126.md`, `results126/report01/SUMMARY126.json`, `results126/primary01/EXECUTION126.json`, `results126/independent01/SUMMARY126.json`, `results126/independent01/policy_pairs126.jsonl`, `provenance126/POLICY_PAIR_RESPONSE_READBACK126.json`.

### B4. The certificate-aware objective fails its intended service connection

**Original request.** It improves residual span in 36 of 36 cells but worsens aggregate rejections and most quadratic cells. The missing theorem is the connection from the trained objective to action-relevant resolution.

**Inherited R125 amendment.** We preserve the R97 result that residual span improved in all 36 cells while aggregate rejections rose from 644 to 757. The new complete-support objective controls a worst compatible original action contrast directly. The acquisition theorem specifies finite feasible support radii, required level and slope precision, a guaranteed optimizer-width contraction and

the independent absolute-value gate. The prospective operational census records strong-target attainment, accepted changes, rejection and stopping outcomes and complete work.

**Current R126 author action.** The new calculation uses original Bellman supports, feasible action contrasts and a regret stopping criterion. The production comparison is specified to retain every candidate, support, refinement, refusal and fallback under matched ordinary qualification and original objective-width requirements. The independent fitting audit evaluates finite VALUE/MARGINAL objectives separately from delivered economic loss.

**Evidence status.** Executed first-campaign evidence and proved components; the stated scientific limits remain open.

The independently executed historical-data audit passed all 56 declared cases under F126. Of 24 VALUE solver calls configured for BVLS, four execute the distinct active-set path and 20 accept the feasible unconstrained solution through the shared initial least-squares path. There are also 24 independently assembled shifted MARGINAL LP solves and eight explicitly matched hinge copies, including four VALUE copies. All 28 VALUE rounded readouts reproduce the originals exactly; 39 of 56 readouts match overall, while all 17 different MARGINAL readouts and their objectives remain recorded. The maximum exact finite-box gap is below 2.34e-4 for VALUE sum squared loss and 2.84e-8 for the new MARGINAL residual. The whole external audit cost is 21.823981663 seconds, with zero economic queries and no deployed actor replaced.

The main and supplement now state and prove finite rational-cut validity, residual correction over the declared box, the original-action regret enclosure and its finite-grid accuracy, and the direct-service projected gap, constrained optimizer bracket, separate regret/value-width gates and strict same-objective policy replacement. General convexity, interpolation and weak-duality ingredients are identified as inherited or classical.

The frozen paid-support campaign executed all 74 pools, 888 contrasts and 5,920 LP calls. Exact independent reconstruction checked 33,331 cuts and 148 candidate regret records. It found 48 strictly sharper contrasts and six sharper regret bounds, but zero newly qualified policy gains and zero changed primary policies. No new economic query was made, and no finite solver optimality was certified. The six improved regret bounds concern capacity actions that remain unqualified, so they are not a delivered economic gain.

Both warm adaptive and direct services met the stronger production regret target in 522 of 522 decisions, versus 296 of 522 for fused control. Direct used 418,957 compiled terminal rows versus 973,410 for warm adaptive, but its complete primary service time was 63.008976 seconds versus 30.825439 seconds. The lower information workload did not produce a first-campaign speed advantage; no fitted residual is used to reverse that conclusion.

The strong warm-adaptive comparator must be separated from fused control. Among 522 direct-versus-warm pairs, the exact independent audit finds 24 strict gains, 26 strict losses, 257 exact policy ties and 215 unresolved orderings; its deterministic equal-weight mean difference is enclosed by [-7.002e-6, 7.125e-6], which does not resolve a mean-cost ordering. Against fused control, 522 pairs yield 214 gains, 13 losses, 260 ties and 35 unresolved orderings, with mean difference enclosed by [-3.681e-4, -3.530e-4]. These are bounds for the fixed census, not population confidence intervals.

The separately source-frozen R126C paired episode executed 3,168 new primary and 1,056 new timed decisions. All 4,224 complete scientific records agree with their original reference after only the prospectively declared clock and run-metadata exclusions. The same within-decision geometry cache is available to every neural and conventional service and is reset before each next paid decision. Complete parent-allocated nine-state costs change from 16.2466 to 15.9305 seconds

for fused control, 35.1063 to 27.4172 for warm adaptive, 68.8494 to 33.5117 for direct terminal, and 7.7973 to 7.7847 for grid DP. Direct cost falls by 51.326 percent but remains 22.229 percent above equally cached warm adaptive. All first-campaign adverse costs remain unchanged. These are descriptive costs, not timing confidence intervals.

**Remaining scientific requirement.** The completed first campaign reports the actual action and full-work effects, including adverse complete-time evidence. A lower residual, a better finite-design optimizer or a bound without production evidence does not establish service improvement.

**Source routes.** `FIT_AUDIT_PROTOCOL126.md`, `code126/independent_fit126.py`, `new126/independent_fit126.tex`, `FIT_AUDIT_REPORT126.md`, `new126/fit_results126.tex`, `new126/fit_development126.tex`, `results126/independent-fit-attempt01/EXECUTION126.json`, `results126/independent-fit-attempt01/SUMMARY126.json`, `results126/launch-fit-attempt01/PROCESS126.json`, `new126/certified_support_main126.tex`, `new126/certified_support_proofs126.tex`, `new126/decision_stopping_main126.tex`, `new126/decision_stopping_proofs126.tex`, `results126/support01/SUMMARY126.json`, `results126/support-audit01/SUPPORT_VERIFICATION126.json`, `new126/support_results126.tex`, `results126/report01/RESULTS126.md`, `results126/report01/SUMMARY126.json`, `results126/primary01/EXECUTION126.json`, `results126/independent01/SUMMARY126.json`, `results126/independent01/policy_pairs126.jsonl`, `provenance126/POLICY_PAIR_RESPONSE_READBACK126.json`, `provenance126/FIT_NUMERICAL_PATH_READBACK126.json`, `SOURCE_FREEZE126C.json`, `BOUNDARIES126C.json`, `new126/cached_envelopes_main126c.tex`, `new126/cached_envelopes_proofs126c.tex`, `results126c/verification01/EXECUTION126.json`, `results126c/report01/SUMMARY126C.json`, `results126c/report01/RESULTS126C.md`.

### B5. Complete work favors a strong conventional method

**Original request.** Quadratic least squares is faster, uses fewer proposal queries, and has no neural acquisition overhead. R97 therefore does not shift the computational frontier toward NBO-specific representation.

**Inherited R125 amendment.** Quadratic least squares remains the recorded R97 complete-work winner. R125 preserves all original acquisition bills and compares seven stored representations, online original-model controls and a 33-action dynamic-programming control specified for fresh original-model evaluation. The four prefix-based services pay their completed strong fused prefixes, rejected candidates, discarded precision levels and fallbacks; the grid control pays its full action scan, refinements and any required fallback. The reporting protocol adds load, serialization, bookkeeping and parent-clock allocation to the recorded service cost.

**Current R126 author action.** The strong original-model fused and direct-terminal conventional services remain in the comparison alongside quadratic/polynomial, spline, grid, ReLU and matched hinge actors. Every original acquisition allocation, paid risk query, rejected candidate, audit and complete process cost stays visible. New independent refitting is charged as research verification, without pretending to have trained the deployed actors again.

**Evidence status.** Executed first-campaign evidence and proved components; the stated scientific limits remain open.

The independently executed historical-data audit passed all 56 declared cases under F126. Of 24 VALUE solver calls configured for BVLS, four execute the distinct active-set path and 20 accept the feasible unconstrained solution through the shared initial least-squares path. There are also 24 independently assembled shifted MARGINAL LP solves and eight explicitly matched hinge copies, including four VALUE copies. All 28 VALUE rounded readouts reproduce the originals exactly; 39 of 56 readouts match overall, while all 17 different MARGINAL readouts and their objectives remain recorded. The maximum exact finite-box gap is below 2.34e-4 for VALUE sum squared loss and 2.84e-8 for the new MARGINAL residual. The whole external audit cost is 21.823981663 seconds, with zero economic queries and no deployed actor replaced.

The frozen paid-support campaign executed all 74 pools, 888 contrasts and 5,920 LP calls. Exact independent reconstruction checked 33,331 cuts and 148 candidate regret records. It found 48 strictly sharper contrasts and six sharper regret bounds, but zero newly qualified policy gains and zero changed primary policies. No new economic query was made, and no finite solver optimality was certified. The six improved regret bounds concern capacity actions that remain unqualified, so they are not a delivered economic gain.

The first campaign preserves the complete conventional comparison: fused control used 14.573211 seconds, warm adaptive 30.825439 seconds, direct 63.008976 seconds, and the 18 action-grid decisions 7.818594 seconds. These groups have their stated counts and stronger-target status, so the clocks are not interpreted as unconditional dominance at unequal accuracy. The direct service performed 1,805 original queries and retained all query/candidate work; there were no direct fallbacks or failures.

The strong warm-adaptive comparator must be separated from fused control. Among 522 direct-versus-warm pairs, the exact independent audit finds 24 strict gains, 26 strict losses, 257 exact policy ties and 215 unresolved orderings; its deterministic equal-weight mean difference is enclosed by [-7.002e-6, 7.125e-6], which does not resolve a mean-cost ordering. Against fused control, 522 pairs yield 214 gains, 13 losses, 260 ties and 35 unresolved orderings, with mean difference enclosed by [-3.681e-4, -3.530e-4]. These are bounds for the fixed census, not population confidence intervals.

The separately source-frozen R126C paired episode executed 3,168 new primary and 1,056 new timed decisions. All 4,224 complete scientific records agree with their original reference after only the prospectively declared clock and run-metadata exclusions. The same within-decision geometry cache is available to every neural and conventional service and is reset before each next paid decision. Complete parent-allocated nine-state costs change from 16.2466 to 15.9305 seconds for fused control, 35.1063 to 27.4172 for warm adaptive, 68.8494 to 33.5117 for direct terminal, and 7.7973 to 7.7847 for grid DP. Direct cost falls by 51.326 percent but remains 22.229 percent above equally cached warm adaptive. All first-campaign adverse costs remain unchanged. These are descriptive costs, not timing confidence intervals.

**Remaining scientific requirement.** The completed first census and cost ledger establish no neural or support-solver frontier improvement. Historical conventional winners remain valid under their original conditions.

**Source routes.** `FIT_AUDIT_PROTOCOL126.md`, `code126/independent_fit126.py`, `new126/independent_fit126.tex`, `FIT_AUDIT_REPORT126.md`, `new126/fit_results126.tex`, `new126/fit_development126.tex`, `results126/independent-fit-attempt01/EXECUTION126.json`, `results126/independent-fit-attempt01/SUMMARY126.json`, `results126/launch-fit-attem`

pt01/PROCESS126.json, results126/support01/SUMMARY126.json, results126/support-audit01/SUPPORT_VERIFICATION126.json, new126/support_results126.tex, results126/report01/RESULTS126.md, results126/report01/SUMMARY126.json, results126/primary01/EXECUTION126.json, results126/independent01/SUMMARY126.json, results126/independent01/policy_pairs126.jsonl, provenance126/POLICY_PAIR_RESPONSE_READBACK126.json, provenance126/FIT_NUMERICAL_PATH_READBACK126.json, SOURCE_FREEZE126C.json, BOUNDARIES126C.json, new126/cached_envelopes_main126c.tex, new126/cached_envelopes_proofs126c.tex, results126c/verification01/EXECUTION126.json, results126c/report01/SUMMARY126C.json, results126c/report01/RESULTS126C.md.

### B6. The purchase exercise is conditional sensitivity, not economic identification

**Original request.** No learned service is adopted over quadratic least squares. Prices and fees are external, clocks are descriptive, and the policy interval is broad. The exercise does not change an economic conclusion.

**Inherited R125 amendment.** The R125 design keeps the same continuous-risk investment primitives, constrained decisions and implemented future controller. Its adoption calculations use the first-state workload for which the timing observations were collected, the full traditional comparator menu, original policy-value intervals and explicit acquisition amortization. Sufficient and optimistic regions are separate. Proposition policyidentity125 proves that equality of complete realized policies implies equality of the original recursive objective; a strict complete-cost ordering then excludes positive compute prices for the more costly service, conditional on the declared timing ranges. This post-execution deduction preserves the frozen price-region classifications.

**Current R126 author action.** Economic evaluation remains under the original continuous-risk investment loss and actually implemented future controller. Conditional price/amortization exercises are explicitly distinguished from observed demand or calibrated adoption. No new experiment outcome substitutes for the inherited zero or unresolved neural-adoption regions.

**Evidence status.** Executed first-campaign evidence and proved components; the stated scientific limits remain open.

The conditional adoption ledger reports 96 of 96 sufficient neural regions empty, none of the 96 optimistic outer regions empty, and 96 exact same-policy conventional cost blockers under recorded allocations and observed timing ranges. Workload and price scenarios are conditional inputs; they do not identify market demand, externally calibrated willingness to pay or general neural nonadoption.

The strong warm-adaptive comparator must be separated from fused control. Among 522 direct-versus-warm pairs, the exact independent audit finds 24 strict gains, 26 strict losses, 257 exact policy ties and 215 unresolved orderings; its deterministic equal-weight mean difference is enclosed by [-7.002e-6, 7.125e-6], which does not resolve a mean-cost ordering. Against fused control, 522 pairs yield 214 gains, 13 losses, 260 ties and 35 unresolved orderings, with mean difference enclosed by [-3.681e-4, -3.530e-4]. These are bounds for the fixed census, not population confidence intervals.

The separately source-frozen R126C paired episode executed 3,168 new primary and 1,056 new timed decisions. All 4,224 complete scientific records agree with their original reference after only the prospectively declared clock and run-metadata exclusions. The same within-decision geometry cache is available to every neural and conventional service and is reset before each next

paid decision. Complete parent-allocated nine-state costs change from 16.2466 to 15.9305 seconds for fused control, 35.1063 to 27.4172 for warm adaptive, 68.8494 to 33.5117 for direct terminal, and 7.7973 to 7.7847 for grid DP. Direct cost falls by 51.326 percent but remains 22.229 percent above equally cached warm adaptive. All first-campaign adverse costs remain unchanged. These are descriptive costs, not timing confidence intervals.

**Remaining scientific requirement.** Externally calibrated losses, compute prices, fees, workload and consequential application evidence remain required for an economic adoption identification claim.

**Source routes.** `results126/report01/RESULTS126.md`, `results126/report01/SUMMARY126.json`, `results126/primary01/EXECUTION126.json`, `results126/independent01/SUMMARY126.json`, `results126/independent01/policy_pairs126.jsonl`, `provenance126/POLICY_PAIR_RESPONSE_READBACK126.json`, `SOURCE_FREEZE126C.json`, `BOUNDARIES126C.json`, `new126/cached_envelopes_main126c.tex`, `new126/cached_envelopes_proofs126c.tex`, `results126c/verification01/EXECUTION126.json`, `results126c/report01/SUMMARY126C.json`, `results126c/report01/RESULTS126C.md`.

### B7. The new 84 actors lack direct population policy-cost inference

**Original request.** R97 transfers only the uniform deterministic qualification bound to the new services. The R96 statistical intervals belong to different frozen actors and cannot establish the new actors' mean-cost ordering.

**Inherited R125 amendment.** The R97 actors retain their own identities; R96 intervals are not transferred to them. The R125 census uses four predetermined dyadic states and reports deterministic original-model bounds for those states. It makes no population-mean claim and creates no confidence interval from timing repetitions. The later R124-B experiment remains a separate protected source/state/execution identity and supplies no substitute outcome to R125.

**Current R126 author action.** The R126 production design uses a declared deterministic off-mesh state census and the exact inherited actors. Its state-level original-policy values and decision losses belong to that census only. No historical R96/R121/R122 population interval is transferred to a different actor, new support solver or selected returned policy.

**Evidence status.** Executed first-campaign evidence and proved components; the stated scientific limits remain open.

The 1,584 primary decisions use nine prospectively fixed dyadic off-mesh states, and the 528 serial timing decisions use their separate first-state workload. There are zero independent population samples. Frozen independent reconstruction confirmed the ordinary regret threshold for all 1,584 decisions, 1,277 reported numerical regret bounds and containment of the independent policy interval inside the reported policy interval in 471 cases; 307 reported-bound confirmations and 1,113 containments remain unresolved, with zero contradictions. Its stronger regret-threshold count is 1,175. Those frozen qualification-named fields check regret only. A separate, explicitly post-execution audit of the saved independent bounds verifies absolute width in all 1,584 cases, and hence complete ordinary qualification in 1,584 and stronger qualification in 1,175 cases. This extra audit makes zero economic evaluations and does not turn the unresolved tighter bounds or

interval containments into confirmations. None of these deterministic-census checks is a population confidence interval.

**Remaining scientific requirement.** A deterministic census supplies no new independent population sample or population confidence interval. New refitted readouts are audit-only and receive no undeclared policy inference.

**Source routes.** `results126/report01/RESULTS126.md`, `results126/report01/SUMMARY126.json`, `results126/primary01/EXECUTION126.json`, `results126/independent01/SUMMARY126.json`, `postexecution126/independent_width_audit126.py`, `results126/independentwidth01/EXECUTION126.json`.

### B8. Independent verification stops before fitting

**Original request.** Feature selection, interval-aware fitting, rounding, domain validation, and serialization are not independently reproduced. This is the critical new acquisition stack.

**Inherited R125 amendment.** R125 freezes and reuses all 56 actual M121 actor files, with their original features and acquisition bills. It performs no new fitting. Its separate rational economic evaluator is specified to audit every distinct final and prefix action using original primitives and a fixed precision ladder and global cap. Independently implemented Fraction diagnostics test the new interpolation construction without importing the production solver.

**Current R126 author action.** The separately frozen and now executed fitting audit recovers all four original label files, all raw fits/designs and all 56 model files from immutable R121 custody. It independently reconstructs fixed feature specifications, exact rational designs and one-sided derivatives; configures BVLS rather than historical TRF for VALUE, with the actual active-set and shared initial-solve paths separately disclosed; independently formulates the shifted-box MARGINAL program; and checks exact dyadic rounding, serialization and rational finite-box objective bounds. The marginal solver still shares HiGHS, which is disclosed. No production fitting or feature implementation is imported.

**Evidence status.** Executed or proved components with explicit remaining scientific scope.

The independently executed historical-data audit passed all 56 declared cases under F126. Of 24 VALUE solver calls configured for BVLS, four execute the distinct active-set path and 20 accept the feasible unconstrained solution through the shared initial least-squares path. There are also 24 independently assembled shifted MARGINAL LP solves and eight explicitly matched hinge copies, including four VALUE copies. All 28 VALUE rounded readouts reproduce the originals exactly; 39 of 56 readouts match overall, while all 17 different MARGINAL readouts and their objectives remain recorded. The maximum exact finite-box gap is below 2.34e-4 for VALUE sum squared loss and 2.84e-8 for the new MARGINAL residual. The whole external audit cost is 21.823981663 seconds, with zero economic queries and no deployed actor replaced.

**Remaining scientific requirement.** This does not independently implement HiGHS, reacquire the economic labels, perform learned hidden-feature selection that R121 never performed, or verify the fitted error over the entire continuous domain. The original labels and production Bellman qualification retain their own separate proofs and evidence.

**Source routes.** FIT_AUDIT_PROTOCOL126.md, code126/independent_fit126.py, new126/independent_fit126.tex, FIT_AUDIT_REPORT126.md, new126/fit_results126.tex, new126/fit_development126.tex, results126/independent-fit-attempt01/EXECUTION126.json, results126/independent-fit-attempt01/SUMMARY126.json, results126/launch-fit-attempt01/PROCESS126.json, provenance126/FIT_NUMERICAL_PATH_READBACK126.json.

### B9. The active submission remains cumulative and disproportionate

**Original request.** A 47-page article, 37-page supplement, and 230-page appendix are excessive for the narrow current theorem and negative acquisition experiment. Historical preservation need not determine the active submission's length.

**Inherited R125 amendment.** The new main text is organized around the original recursive economic problem, the complete-support information theorem, its acquisition implications and the current prospective comparison. The full mathematical supplement and development companion retain the earlier proofs, corrections, failures and negative outcomes. A new source census pins all 366 original branch heads and separates the complete R122 manuscript from later R123/R124 research states.

**Current R126 author action.** The current article centers the original economic problem, its Bellman construction and the new action-level calculation. Full proofs stay in the supplement. All historical source, negative results, failed attempts and report identities are preserved through the development companion and exact retention map.

**Evidence status.** Author organization and source-preservation routes are concrete; actual package receipts identify the compiled documents, without a remote-CI claim here.

The four current author wrappers route the original Bellman problem, action-relevant methods and executed economic comparisons into the main article, complete proofs into the supplement, and historical development into the companion. The retention audit preserves 325 explicitly mapped source records, exact prior wrappers and all 1,135 tracked R125 files byte-for-byte. The actual native build receipts identify compiled inputs and documents; SOURCE_PDF_HASHES126.json and REFEREE_MANIFEST126.json bind the final source/PDF and delivery closure. REPRODUCE126.md provides the concrete commands. This response makes no unobserved remote-CI success claim.

**Remaining scientific requirement.** Scientific contribution is assessed by the new theorem/algorithm/economic evidence, not by the accumulated number of revisions, preserved pages or verified hashes.

**Source routes.** ECTA.tex, supp.tex, appendix.tex, RESPONSE_TO_REFEREE.tex, CONTENT_RETENTION126.json, CONTENT_RETENTION126.md, REFEREE_MANIFEST126.json, SOURCE_PDF_HASHES126.json, REPRODUCE126.md, provenance126/R125_REPOSITORY_CUSTODY126.json, provenance126/CURRENT_RETENTION_CHECK126.json.

### B10. The negative result is more compelling than the platform claim

**Original request.** The strongest new fact is that a valid global residual objective can fail to improve delivered decisions. The manuscript should either make that misalignment central or

supply a new action-aligned objective and theorem.

**Inherited R125 amendment.** The residual-versus-decision mismatch determines the revision priority. We add an exact action-contrast information program and a finite support-precision/work theorem. The latter distinguishes optimizer localization, original regret, absolute policy-value width and strict same-future policy improvement. This prevents a reduction in internal fit or optimizer width from being called a delivered economic gain.

**Current R126 author action.** The inherited misalignment is treated as a scientific constraint. The current intervention acts on original-action information and regret stopping, and tests delivered original decisions and complete work rather than selecting a fit by its internal residual alone. The acquisition audit explicitly separates finite-design objective checks from policy certification.

**Evidence status.** Executed first-campaign evidence and proved components; the stated scientific limits remain open.

The independently executed historical-data audit passed all 56 declared cases under F126. Of 24 VALUE solver calls configured for BVLS, four execute the distinct active-set path and 20 accept the feasible unconstrained solution through the shared initial least-squares path. There are also 24 independently assembled shifted MARGINAL LP solves and eight explicitly matched hinge copies, including four VALUE copies. All 28 VALUE rounded readouts reproduce the originals exactly; 39 of 56 readouts match overall, while all 17 different MARGINAL readouts and their objectives remain recorded. The maximum exact finite-box gap is below 2.34e-4 for VALUE sum squared loss and 2.84e-8 for the new MARGINAL residual. The whole external audit cost is 21.823981663 seconds, with zero economic queries and no deployed actor replaced.

The main and supplement now state and prove finite rational-cut validity, residual correction over the declared box, the original-action regret enclosure and its finite-grid accuracy, and the direct-service projected gap, constrained optimizer bracket, separate regret/value-width gates and strict same-objective policy replacement. General convexity, interpolation and weak-duality ingredients are identified as inherited or classical.

The frozen paid-support campaign executed all 74 pools, 888 contrasts and 5,920 LP calls. Exact independent reconstruction checked 33,331 cuts and 148 candidate regret records. It found 48 strictly sharper contrasts and six sharper regret bounds, but zero newly qualified policy gains and zero changed primary policies. No new economic query was made, and no finite solver optimality was certified. The six improved regret bounds concern capacity actions that remain unqualified, so they are not a delivered economic gain.

The direct stopping rule meets its two distinct original-objective gates. The first implementation nonetheless more than doubles warm-adaptive primary service time while reducing compiled terminal rows by about 57 percent. The complete-support calculation sharpens some certificates but generates zero new qualified policy gains. Both negative results remain explicit; neither is presented as completion of the original NBO research objective.

**Remaining scientific requirement.** A favorable fit objective is not completion of the original NBO objective. Comparative action and economic gains must follow from the actual new theorem and executed experiment.

**Source routes.** `FIT_AUDIT_PROTOCOL126.md`, `code126/independent_fit126.py`, `new126/independent_fit126.tex`, `FIT_AUDIT_REPORT126.md`, `new126/fit_results126.tex`, `new126/fit_devel`

opment126.tex, results126/independent-fit-attempt01/EXECUTION126.json, results126/independent-fit-attempt01/SUMMARY126.json, results126/launch-fit-attempt01/PROCESS126.json, new126/certified_support_main126.tex, new126/certified_support_proofs126.tex, new126/decision_stopping_main126.tex, new126/decision_stopping_proofs126.tex, results126/support01/SUMMARY126.json, results126/support-audit01/SUPPORT_VERIFICATION126.json, new126/support_results126.tex, results126/report01/RESULTS126.md, results126/report01/SUMMARY126.json, results126/primary01/EXECUTION126.json, results126/independent01/SUMMARY126.json, provenance126/FIT_NUMERICAL_PATH_READBACK126.json.

### M1. Recast the paper around one method-neutral question

**Original request.** Ask when a stored continuation can be safely purchased for recursive-risk dynamic decisions. Treat neural networks as one optional representation, not the source of validity.

**Inherited R125 amendment.** The manuscript remains Neural Bellman Operators and retains the Econometrica numerical-methods aim. The economic question is how learned continuation values and feasible actor proposals can deliver better original recursive decisions with defensible complete costs. The new mathematics isolates the information and certification operations shared with conventional representations, so the neural contribution is tested rather than assumed.

**Current R126 author action.** We implement the requested decision criterion inside the original Neural Bellman Operators programme. The paper continues to construct, learn and compare recursive Bellman operators, continuation critics and feasible actors in the original decision problem. Shared validity is stated explicitly while neural representation remains a substantive object of investigation.

**Evidence status.** Executed first-campaign evidence and proved components; the stated scientific limits remain open.

The main and supplement now state and prove finite rational-cut validity, residual correction over the declared box, the original-action regret enclosure and its finite-grid accuracy, and the direct-service projected gap, constrained optimizer bracket, separate regret/value-width gates and strict same-objective policy replacement. General convexity, interpolation and weak-duality ingredients are identified as inherited or classical.

The completed production study evaluates the original recursive-risk investment loss with unchanged constraints, risk laws, 56 actor files and future controller. Neural and conventional critics enter the same original-action certificates and strict replacement rule. This is an author revision of the original NBO programme with its current comparative limits made explicit.

**Remaining scientific requirement.** This author revision retains the original title, research identity and Econometrica objective. The scientific objections are answered within that programme; a change of topic is not a substitute for resolving them.

**Source routes.** new126/certified_support_main126.tex, new126/certified_support_proofs126.tex, new126/decision_stopping_main126.tex, new126/decision_stopping_proofs126.tex, results126/report01/RESULTS126.md, results126/report01/SUMMARY126.json, results126/primary01/EXECUTION126.json, results126/independent01/SUMMARY126.json.

### M2. Establish the true novelty of the information theorem

**Original request.** Give a theorem-by-theorem comparison with information-based complexity. Add a genuinely new randomized, nonuniform-law, multivariate, or Bellman-compositional result, or present the current theorem as a specialized supporting lemma.

**Inherited R125 amendment.** The new proof identifies the specific extension beyond smooth convex interpolation and R124: all original quadratic supports are retained through exact node-conjugate inequalities and a constructive global interpolant. It distinguishes the shared-slope lifting from the equivalent full pointwise robust obstacle. The R123 midpoint theorem, the R124 residual certificate and the inherited fixed-alpha value-oracle rate remain attributed to their prior sources.

**Current R126 author action.** The novelty map distinguishes exact information supplied by all paid supports, the finite rational-cut calculation and its decision stopping condition from classical convex interpolation, weak duality and the older deterministic AVaR exponent. Finite-design audit duality is explicitly presented as a verification lemma.

**Evidence status.** Executed or proved components with explicit remaining scientific scope.

The main and supplement now state and prove finite rational-cut validity, residual correction over the declared box, the original-action regret enclosure and its finite-grid accuracy, and the direct-service projected gap, constrained optimizer bracket, separate regret/value-width gates and strict same-objective policy replacement. General convexity, interpolation and weak-duality ingredients are identified as inherited or classical.

The frozen paid-support campaign executed all 74 pools, 888 contrasts and 5,920 LP calls. Exact independent reconstruction checked 33,331 cuts and 148 candidate regret records. It found 48 strictly sharper contrasts and six sharper regret bounds, but zero newly qualified policy gains and zero changed primary policies. No new economic query was made, and no finite solver optimality was certified. The six improved regret bounds concern capacity actions that remain unqualified, so they are not a delivered economic gain.

The separately source-frozen R126C paired episode executed 3,168 new primary and 1,056 new timed decisions. All 4,224 complete scientific records agree with their original reference after only the prospectively declared clock and run-metadata exclusions. The same within-decision geometry cache is available to every neural and conventional service and is reset before each next paid decision. Complete parent-allocated nine-state costs change from 16.2466 to 15.9305 seconds for fused control, 35.1063 to 27.4172 for warm adaptive, 68.8494 to 33.5117 for direct terminal, and 7.7973 to 7.7847 for grid DP. Direct cost falls by 51.326 percent but remains 22.229 percent above equally cached warm adaptive. All first-campaign adverse costs remain unchanged. These are descriptive costs, not timing confidence intervals.

**Remaining scientific requirement.** The fixed-alpha point-value rate is not a broad randomized or multivariate theorem. Any broader lower/upper bound must specify a common model class and information interface.

**Source routes.** `new126/certified_support_main126.tex`, `new126/certified_support_proofs126.tex`, `new126/decision_stopping_main126.tex`, `new126/decision_stopping_proofs126.tex`, `results126/support01/SUMMARY126.json`, `results126/support-audit01/SUPPORT_VERIFICATION126.json`, `new126/support_results126.tex`, `SOURCE_FREEZE126C.json`, `BOUNDARIES126C.json`,

new126/cached_envelopes_main126c.tex, new126/cached_envelopes_proofs126c.tex, results126c/verification01/EXECUTION126.json, results126c/report01/SUMMARY126C.json, results126c/report01/RESULTS126C.md.

### M3. Develop an action-aligned acquisition objective

**Original request.** Construct a loss that controls rejection, endpoint work, or policy-cost resolution on a predeclared context law. Prove the connection and validate it prospectively against complete work.

**Inherited R125 amendment.** The new action objective is the largest compatible original contrast, and its support-complete feasible set is exact. The acquisition theorem gives a finite-radius precision requirement and, conditional on a separately verified oracle-cost bound, an optimal guaranteed-width schedule for that declared planning problem. The frozen census compares warm adaptive, cold adaptive and fixed-precision acquisition against a fully paid strong fused control, with a stronger original-regret target stated in advance.

**Current R126 author action.** The production algorithm targets original-action resolution using the complete stored support information and a regret stopping certificate. The fixed comparison is designed before new evaluations and retains all refusals and work. A newly solved finite training objective remains a separate audit object and is not silently used to choose the deployed policy.

**Evidence status.** Executed first-campaign evidence and proved components; the stated scientific limits remain open.

The main and supplement now state and prove finite rational-cut validity, residual correction over the declared box, the original-action regret enclosure and its finite-grid accuracy, and the direct-service projected gap, constrained optimizer bracket, separate regret/value-width gates and strict same-objective policy replacement. General convexity, interpolation and weak-duality ingredients are identified as inherited or classical.

The frozen paid-support campaign executed all 74 pools, 888 contrasts and 5,920 LP calls. Exact independent reconstruction checked 33,331 cuts and 148 candidate regret records. It found 48 strictly sharper contrasts and six sharper regret bounds, but zero newly qualified policy gains and zero changed primary policies. No new economic query was made, and no finite solver optimality was certified. The six improved regret bounds concern capacity actions that remain unqualified, so they are not a delivered economic gain.

The original-action stopping implementation was source-frozen before its new evaluations and executes 1,584 predeclared decisions. Direct and warm adaptive both meet the stronger production target for all 522 respective cases, but the first direct implementation is slower despite using fewer terminal rows. Its 689 strict internal replacements have the declared same-objective certificate; direct-versus-control comparisons separately retain 39 independently certified losses. The acquired actors and their training objective are not retrospectively selected by those results.

The strong warm-adaptive comparator must be separated from fused control. Among 522 direct-versus-warm pairs, the exact independent audit finds 24 strict gains, 26 strict losses, 257 exact policy ties and 215 unresolved orderings; its deterministic equal-weight mean difference is enclosed by [-7.002e-6, 7.125e-6], which does not resolve a mean-cost ordering. Against fused control, 522 pairs yield 214 gains, 13 losses, 260 ties and 35 unresolved orderings, with mean difference enclosed by [-3.681e-4, -3.530e-4]. These are bounds for the fixed census, not population confidence intervals.

The separately source-frozen R126C paired episode executed 3,168 new primary and 1,056 new timed decisions. All 4,224 complete scientific records agree with their original reference after only the prospectively declared clock and run-metadata exclusions. The same within-decision geometry cache is available to every neural and conventional service and is reset before each next paid decision. Complete parent-allocated nine-state costs change from 16.2466 to 15.9305 seconds for fused control, 35.1063 to 27.4172 for warm adaptive, 68.8494 to 33.5117 for direct terminal, and 7.7973 to 7.7847 for grid DP. Direct cost falls by 51.326 percent but remains 22.229 percent above equally cached warm adaptive. All first-campaign adverse costs remain unchanged. These are descriptive costs, not timing confidence intervals.

**Remaining scientific requirement.** The connection must be demonstrated in the exact assumptions, implementation and complete executed comparison; merely minimizing an internal objective does not close the request.

**Source routes.** `new126/certified_support_main126.tex`, `new126/certified_support_proofs126.tex`, `new126/decision_stopping_main126.tex`, `new126/decision_stopping_proofs126.tex`, `results126/support01/SUMMARY126.json`, `results126/support-audit01/SUPPORT_VERIFICATION126.json`, `new126/support_results126.tex`, `results126/report01/RESULTS126.md`, `results126/report01/SUMMARY126.json`, `results126/primary01/EXECUTION126.json`, `results126/independent01/SUMMARY126.json`, `results126/independent01/policy_pairs126.jsonl`, `provenance126/POLICY_PAIR_RESPONSE_READBACK126.json`, `SOURCE_FREEZE126C.json`, `BOUNDARIES126C.json`, `new126/cached_envelopes_main126c.tex`, `new126/cached_envelopes_proofs126c.tex`, `results126c/verification01/EXECUTION126.json`, `results126c/report01/SUMMARY126C.json`, `results126c/report01/RESULTS126C.md`.

### M4. Expand the conventional geometry frontier

**Original request.** Compare ReLU with conventional oblique hinges, adaptive triangulation, convex regression, and representation-neutral basis selection under equal information, rank, acquisition, and validation budgets.

**Inherited R125 amendment.** R125 retains quadratic, degree-two and degree-three polynomial, spline, grid, ReLU and the exact ordinary-hinge realization, together with online controls and a 33-action original-model DP control. Every stored actor receives the same four service rules and target gates. The matched hinge inherits the actual parent ReLU acquisition cost where required, so equal geometry does not create a fictitious free comparator or a fictitious neural feature advantage.

**Current R126 author action.** The retained representation frontier includes the exact ordinary-hinge encoding, signed and constrained quadratic/polynomial families, cubic spline, grid and strong analytic/semi-analytic original-model controls. Both losses see the same original labels, boxes and declared features. The new audit reconstructs these feature specifications independently.

**Evidence status.** Executed first-campaign evidence and proved components; the stated scientific limits remain open.

The independently executed historical-data audit passed all 56 declared cases under F126. Of 24 VALUE solver calls configured for BVLS, four execute the distinct active-set path and 20 accept the feasible unconstrained solution through the shared initial least-squares path. There are also 24

independently assembled shifted MARGINAL LP solves and eight explicitly matched hinge copies, including four VALUE copies. All 28 VALUE rounded readouts reproduce the originals exactly; 39 of 56 readouts match overall, while all 17 different MARGINAL readouts and their objectives remain recorded. The maximum exact finite-box gap is below 2.34e-4 for VALUE sum squared loss and 2.84e-8 for the new MARGINAL residual. The whole external audit cost is 21.823981663 seconds, with zero economic queries and no deployed actor replaced.

The executed catalogue includes all seven fixed-feature representations, both losses, both acquisition mesh sizes and both risk levels, together with online and action-grid controls. Exact hinge matches remain charged for their parent ReLU acquisition. The independent fitting audit reconstructs every fixed feature specification and all 56 readout cases. Adaptive triangulation, convex regression and learned feature selection remain unexecuted comparators, as explicitly stated.

The C episode supplies 192 sufficient neural adoption regions, all empty, with 192 exact same-policy conventional cost blockers; all 192 optimistic outer regions remain nonempty. Exact reuse improves common certificate computation without establishing a neural-specific representation, information or policy advantage.

**Remaining scientific requirement.** The present audit does not add a learned feature-selection, adaptive-triangulation or convex-regression tournament. Such methods cannot be claimed as defeated without matched implementation and resource accounting.

**Source routes.** `FIT_AUDIT_PROTOCOL126.md`, `code126/independent_fit126.py`, `new126/independent_fit126.tex`, `FIT_AUDIT_REPORT126.md`, `new126/fit_results126.tex`, `new126/fit_development126.tex`, `results126/independent-fit-attempt01/EXECUTION126.json`, `results126/independent-fit-attempt01/SUMMARY126.json`, `results126/launch-fit-attempt01/PROCESS126.json`, `results126/report01/RESULTS126.md`, `results126/report01/SUMMARY126.json`, `results126/primary01/EXECUTION126.json`, `results126/independent01/SUMMARY126.json`, `provenance126/FIT_NUMERICAL_PATH_READBACK126.json`, `SOURCE_FREEZE126C.json`, `BOUNDARIES126C.json`, `new126/cached_envelopes_main126c.tex`, `new126/cached_envelopes_proofs126c.tex`, `results126c/verification01/EXECUTION126.json`, `results126c/report01/SUMMARY126C.json`, `results126c/report01/RESULTS126C.md`.

### M5. Treat the negative result as first-class evidence

**Original request.** Report aggregate and cell-level rejection changes beside span improvements, explain where the global objective allocates approximation effort, and do not label 36/36 span reduction as delivered-service success.

**Inherited R125 amendment.** R97's 36/36 span reductions, rejection increase, quadratic work victory and zero adopted neural regions remain visible and source-bound. Later negative acquisition and full-menu outcomes remain attached to their own campaign identities. Current reporting separates accepted same-incumbent gains, stronger-target attainment, action errors, service work and comparative purchase conditions.

**Current R126 author action.** The R97 result is retained explicitly: all 36 certificate-aware pairs reduced the validated residual span, while total first-proposal rejections rose from 644 to 757;

quadratic least squares remained the full-work winner. The new report presents action/refusal outcomes beside certificate quantities and charges all failed candidates.

**Evidence status.** Executed first-campaign evidence and proved components; the stated scientific limits remain open.

The first new campaign retains 238 independently certified direct-versus-control improvements and 39 certified losses, as well as 517 exact ties and 250 unresolved pairs. It also retains the adverse complete-time comparison despite fewer terminal rows. These are distinct from the preserved R97 residual/rejection experiment; a favorable new metric cannot overwrite the old rejection increase.

The C episode supplies 192 sufficient neural adoption regions, all empty, with 192 exact same-policy conventional cost blockers; all 192 optimistic outer regions remain nonempty. Exact reuse improves common certificate computation without establishing a neural-specific representation, information or policy advantage.

**Remaining scientific requirement.** Historical adverse outcomes retain their original source and conditions. New favorable cells, if any, cannot erase them or change the old evaluation rule.

**Source routes.** `results126/report01/RESULTS126.md`, `results126/report01/SUMMARY126.json`, `results126/primary01/EXECUTION126.json`, `results126/independent01/SUMMARY126.json`, `SOURCE_FREEZE126C.json`, `BOUNDARIES126C.json`, `new126/cached_envelopes_main126c.tex`, `new126/cached_envelopes_proofs126c.tex`, `results126c/verification01/EXECUTION126.json`, `results126c/report01/SUMMARY126C.json`, `results126c/report01/RESULTS126C.md`.

### M6. Execute a consequential application or narrow the claim

**Original request.** Either calibrate policy margins, compute prices, fees, and workloads in an application where adoption matters, or state that the paper is about numerical certification rather than economic purchase.

**Inherited R125 amendment.** The investment economy remains an actual constrained recursive decision problem with continuous innovations, nested AVaR, original objective values, regret and computation bills. The new theorem explains when original information can license an economically meaningful action change, and the protocol keeps monetary computation scenarios separate from observed policy-value bounds. The post-execution Proposition policyidentity125 couples equal complete policies before comparing their costs, rather than treating their objective intervals as independent unknown values.

**Current R126 author action.** The paper keeps the original recursive investment economy and its neural Bellman research objective. It reports original objective bounds, regret, feasible action changes, computational costs and conditional adoption implications in their correct units. It does not relabel a sensitivity grid as a calibrated market adoption result.

**Evidence status.** Executed first-campaign evidence and proved components; the stated scientific limits remain open.

The measured original-model policy comparisons have economic meaning within their declared loss and state census: 238 strict direct gains, 39 strict losses, 517 exact ties and 250 unresolved pairs

against the two completed controls. Conditional adoption still gives no sufficient neural region. These observations are not a calibrated substantive market adoption exercise and do not complete that empirical requirement.

The strong warm-adaptive comparator must be separated from fused control. Among 522 direct-versus-warm pairs, the exact independent audit finds 24 strict gains, 26 strict losses, 257 exact policy ties and 215 unresolved orderings; its deterministic equal-weight mean difference is enclosed by [-7.002e-6, 7.125e-6], which does not resolve a mean-cost ordering. Against fused control, 522 pairs yield 214 gains, 13 losses, 260 ties and 35 unresolved orderings, with mean difference enclosed by [-3.681e-4, -3.530e-4]. These are bounds for the fixed census, not population confidence intervals.

The C cache has exactly the same complete policies, values, regrets and economic work as its uncached reference. Its cost reduction is a computational result; it creates neither a new policy improvement nor a calibrated application observation.

**Remaining scientific requirement.** A consequential calibrated economic application remains a distinct scientific requirement. The author retains the broader research programme and venue objective while recording this incomplete empirical claim honestly.

**Source routes.** `results126/report01/RESULTS126.md`, `results126/report01/SUMMARY126.json`, `results126/primary01/EXECUTION126.json`, `results126/independent01/SUMMARY126.json`, `results126/independent01/policy_pairs126.jsonl`, `provenance126/POLICY_PAIR_RESPONSE_READBACK126.json`, `SOURCE_FREEZE126C.json`, `BOUNDARIES126C.json`, `new126/cached_envelopes_main126c.tex`, `new126/cached_envelopes_proofs126c.tex`, `results126c/verification01/EXECUTION126.json`, `results126c/report01/SUMMARY126C.json`, `results126c/report01/RESULTS126C.md`.

### M7. Add a genuine scaling experiment

**Original request.** Vary horizon, controls, innovation dimension, and risk level under matched comparators. Report product-point counts, memory, endpoint work, acquisition, and complete service work.

**Inherited R125 amendment.** The R125 study varies representation, original risk level, information budget and service precision strategy within the same economy. It records requested terminal contexts, actual operation counters, all refinement and fallback work, and the implemented stopping outcomes. The theory distinguishes scalar action information from unrestricted Bellman complexity.

**Current R126 author action.** The new design changes the declared off-mesh initial-state census and decision algorithms within the same original economy. The implementation records actual support counts, risk resolution, descendant work, memory and complete costs.

**Evidence status.** Executed first-campaign evidence and proved components; the stated scientific limits remain open.

The current executed workload consists of 176 service cohorts and 1,584 primary decisions, plus 528 serial timing decisions. The operation ledger reports 902,961 compiled terminal rows for fused control, 973,410 for warm adaptive, 418,957 for direct, and 176,223 for the grid baseline. These are actual same-economy workloads, not a claimed horizon/control/innovation-dimension scaling study.

**Remaining scientific requirement.** The state census is not a new horizon/control/innovation-dimension scaling experiment. The two-date, two-dimensional scalar-control limits remain explicit, as do adverse inherited horizon results.

**Source routes.** `results126/report01/RESULTS126.md`, `results126/report01/SUMMARY126.json`, `results126/primary01/EXECUTION126.json`, `results126/independent01/SUMMARY126.json`.

### M8. Strengthen timing evidence

**Original request.** Use more repetitions, record or control processor state, and make operation counts primary. Predeclare the inferential target for deployment clocks if a probabilistic purchase claim is desired.

**Inherited R125 amendment.** The frozen protocol specifies exactly three serial timing processes, all 234 service cohorts on the same first state, deterministic cohort-order rotation and one-thread numerical settings. It requires action, certificate, stopping and operation identity against the primary first-state decisions, retains every timing, records parent and child clocks and charges parent residuals.

**Current R126 author action.** Timing is attached to the actual source, actor and state identities and uses the prospectively declared repetitions. Operation/work counts remain primary. The independent acquisition audit is separately clocked as research work and cannot be counted as a fresh deployment timing sample.

**Evidence status.** Executed first-campaign evidence and proved components; the stated scientific limits remain open.

The independently executed historical-data audit passed all 56 declared cases under F126. Of 24 VALUE solver calls configured for BVLS, four execute the distinct active-set path and 20 accept the feasible unconstrained solution through the shared initial least-squares path. There are also 24 independently assembled shifted MARGINAL LP solves and eight explicitly matched hinge copies, including four VALUE copies. All 28 VALUE rounded readouts reproduce the originals exactly; 39 of 56 readouts match overall, while all 17 different MARGINAL readouts and their objectives remain recorded. The maximum exact finite-box gap is below 2.34e-4 for VALUE sum squared loss and 2.84e-8 for the new MARGINAL residual. The whole external audit cost is 21.823981663 seconds, with zero economic queries and no deployed actor replaced.

Three serial complete first-state timing repetitions, each with 176 decisions, have actual exit-zero receipts. The first nine-state primary campaign and every observed timing range remain preserved. Frequency/load were not controlled, so clocks remain descriptive. The later fitting audit was deliberately run after those serial measurements and is separately charged shared-host research work.

The separately source-frozen R126C paired episode executed 3,168 new primary and 1,056 new timed decisions. All 4,224 complete scientific records agree with their original reference after only the prospectively declared clock and run-metadata exclusions. The same within-decision geometry cache is available to every neural and conventional service and is reset before each next paid decision. Complete parent-allocated nine-state costs change from 16.2466 to 15.9305 seconds for fused control, 35.1063 to 27.4172 for warm adaptive, 68.8494 to 33.5117 for direct terminal, and 7.7973 to 7.7847 for grid DP. Direct cost falls by 51.326 percent but remains 22.229 percent above

equally cached warm adaptive. All first-campaign adverse costs remain unchanged. These are descriptive costs, not timing confidence intervals.

For 176 paired first-state cohorts, cache-versus-uncached observed ranges show one strict cache win, one strict loss and 174 overlaps. The pooled nine-state reduction is therefore not generalized into per-cohort or hardware-population dominance.

**Remaining scientific requirement.** Same-host repetition ranges are descriptive, not hardware-population confidence intervals. No new CPU-frequency control, cross-machine generalization or probabilistic deployment-price claim is inferred.

**Source routes.** `FIT_AUDIT_PROTOCOL126.md`, `code126/independent_fit126.py`, `new126/independent_fit126.tex`, `FIT_AUDIT_REPORT126.md`, `new126/fit_results126.tex`, `new126/fit_development126.tex`, `results126/independent-fit-attempt01/EXECUTION126.json`, `results126/independent-fit-attempt01/SUMMARY126.json`, `results126/launch-fit-attempt01/PROCESS126.json`, `results126/report01/RESULTS126.md`, `results126/report01/SUMMARY126.json`, `results126/primary01/EXECUTION126.json`, `results126/independent01/SUMMARY126.json`, `provenance126/FIT_NUMERICAL_PATH_READBACK126.json`, `SOURCE_FREEZE126C.json`, `BOUNDARIES126C.json`, `new126/cached_envelopes_main126c.tex`, `new126/cached_envelopes_proofs126c.tex`, `results126c/verification01/EXECUTION126.json`, `results126c/report01/SUMMARY126C.json`, `results126c/report01/RESULTS126C.md`.

### M9. Independently reimplement fitting and package a standalone environment

**Original request.** Reproduce feature selection, interval-aware fitting, rounding, whole-domain validation, and serialization independently. Include a container or lockfile and all inherited modules needed to run the new tests from the artifact alone.

**Inherited R125 amendment.** The prospective R125 source manifest includes scientific code, public frozen dependencies and all 56 actor files. A remote source readback gate precedes economics. The independent evaluator uses the separately implemented rational original primitives; replay, mathematical fixtures, actual economics, source verification and CI each retain their own status and receipt.

**Current R126 author action.** The complete independent fitting closure is vendored and hashed: 181 immutable input files, all 56 saved actors, independent feature/design assembly and refitting code, exact finite-box certificate checks, protocol and reproducible commands. VALUE configures a different active-set algorithm, which actually iterates in four cases while 20 return through the shared unconstrained initial solve; MARGINAL assembly is independent but uses the shared HiGHS library. Whole-domain and policy checks remain explicitly separate.

**Evidence status.** Executed finite-design audit and fresh offline source-gate verification; all stated independence and scientific-scope limits retained.

The independently executed historical-data audit passed all 56 declared cases under F126. Of 24 VALUE solver calls configured for BVLS, four execute the distinct active-set path and 20 accept the feasible unconstrained solution through the shared initial least-squares path. There are also 24 independently assembled shifted MARGINAL LP solves and eight explicitly matched hinge copies, including four VALUE copies. All 28 VALUE rounded readouts reproduce the originals exactly;

39 of 56 readouts match overall, while all 17 different MARGINAL readouts and their objectives remain recorded. The maximum exact finite-box gap is below 2.34e-4 for VALUE sum squared loss and 2.84e-8 for the new MARGINAL residual. The whole external audit cost is 21.823981663 seconds, with zero economic queries and no deployed actor replaced.

A minimal offline Git witness contains the authentic F126 commit, 67 required path trees and 148 distinct blobs serving all 184 gate paths. The 1,834,756-byte pack was installed in a fresh remote-free repository and installed again without changing refs, HEAD or the worktree index. Both installations verified every gated Git file; the unchanged frozen source_gate then verified all 184 reads and 365 assertions before numerical imports. This is verified offline source identity, not a full repository clone or a new fit. Pinned requirements and REPRODUCE126.md complete the runnable archive instructions; native build and source/PDF identity are bound by their actual final receipts.

**Remaining scientific requirement.** All 56 fitting cases and the fresh offline source-identity bootstrap actually executed. Shared initial least-squares and HiGHS infrastructure, unreacquired labels and finite-design scope remain disclosed. The archive does not claim to reimplement every numerical library, validate fitted approximation over the whole continuous domain, or complete every historical experiment.

**Source routes.** `FIT_AUDIT_PROTOCOL126.md`, `code126/independent_fit126.py`, `new126/independent_fit126.tex`, `FIT_AUDIT_REPORT126.md`, `new126/fit_results126.tex`, `new126/fit_development126.tex`, `results126/independent-fit-attempt01/EXECUTION126.json`, `results126/independent-fit-attempt01/SUMMARY126.json`, `results126/launch-fit-attempt01/PROCESS126.json`, `provenance126/FIT_NUMERICAL_PATH_READBACK126.json`, `offline_git126/README.md`, `offline_git126/bootstrap126.py`, `offline_git126/WITNESS_MANIFEST126.json`, `offline_git126/OFFLINE_VERIFICATION126.json`, `offline_git126/FROZEN_GATE_OFFLINE_READBACK126.json`, `requirements126.txt`, `REPRODUCE126.md`, `SOURCE_PDF_HASHES126.json`, `REFEREE_MANIFEST126.json`.

### M10. Shorten the active submission substantially

**Original request.** Move historical material to an archival companion. The active article should contain only the current qualification theorem, precise information result, acquisition objective, matched experiment, and supported economic interpretation.

**Inherited R125 amendment.** The active article presents the economic operator, current complete-support and acquisition results, the matched original-model comparison and supported economic interpretation. The supplement contains complete proofs; the development companion retains historical material, including the original 620 major-response blocks and additional inherited item responses. All 35 genuine report bodies and their manifests are preserved.

**Current R126 author action.** The main article follows one original-economic chain: Bellman operator and learned continuations, action-relevant information, the new feasible decision calculation, proofs, matched implementation and measured economic consequences. Historical derivations and diagnostics remain in a traceable development companion.

**Evidence status.** Author organization and source-preservation routes are concrete; actual package receipts identify the compiled documents, without a remote-CI claim here.

The four current author wrappers route the original Bellman problem, action-relevant methods and executed economic comparisons into the main article, complete proofs into the supplement, and historical development into the companion. The retention audit preserves 325 explicitly mapped source records, exact prior wrappers and all 1,135 tracked R125 files byte-for-byte. The actual native build receipts identify compiled inputs and documents; SOURCE_PDF_HASHES126.json and REFEREE_MANIFEST126.json bind the final source/PDF and delivery closure. REPRODUCE126.md provides the concrete commands. This response makes no unobserved remote-CI success claim.

**Remaining scientific requirement.** Presentation changes do not close a scientific deficit. Every new headline must be supported by a proof and the actual scope of its executed evidence.

**Source routes.** `ECTA.tex`, `supp.tex`, `appendix.tex`, `RESPONSE_TO_REFEREE.tex`, `CONTENT_RETENTION126.json`, `CONTENT_RETENTION126.md`, `REFEREE_MANIFEST126.json`, `SOURCE_PDF_HASHES126.json`, `REPRODUCE126.md`, `provenance126/R125_REPOSITORY_CUSTODY126.json`, `provenance126/CURRENT_RETENTION_CHECK126.json`.

### Minor 1. State 'N^{-2}' consistently as an error-versus-information rate; "information exponent two" is easy to misread.

**Original request.** 1. State 'N^{-2}' consistently as an error-versus-information rate; "information exponent two" is easy to misread.

**Inherited R125 amendment.** The inherited N^{-2} statement is described as an error-versus-information rate. The new complete-support program instead gives an exact finite-record contrast bound, and the work theorem has a conditional exponent p fixed by its stated oracle premise.

**Current R126 author action.** The inherited N^{-2} statement is presented as an error-versus-information rate. The new finite-support and finite-design results use their own information classes and are not renamed information exponent results.

**Evidence status.** Source bound textual revision.

**Remaining scientific requirement.** No new tail-uniform or multivariate rate follows from this terminology correction.

### Minor 2. Put "deterministic, point-value, univariate, fixed-alpha" adjacent to every high-level summary of the lower bound.

**Original request.** 2. Put "deterministic, point-value, univariate, fixed-alpha" adjacent to every high-level summary of the lower bound.

**Inherited R125 amendment.** Every current summary of the inherited lower-bound rate retains its deterministic, point-value, univariate, fixed-alpha qualifications. The separate new randomized-action bracket statement gives its own information class explicitly.

**Current R126 author action.** High-level discussion retains the qualifiers deterministic, point-value, univariate, uniform-law and fixed-alpha for the inherited AVaR lower bound.

**Evidence status.** Source bound textual revision.

**Remaining scientific requirement.** The qualifier set cannot be omitted when comparing the result with broader recursive-risk computation.

### Minor 3. Discuss the opposite tail-level behavior of the lower and upper constants as 'alpha' approaches one.

**Original request.** 3. Discuss the opposite tail-level behavior of the lower and upper constants as 'alpha' approaches one.

**Inherited R125 amendment.** The inherited fixed-alpha result and its differing upper and lower tail constants remain in the retained information discussion. R125 continues to use the two predeclared original risk levels $1/2$ and $3/4$ and does not infer behavior as alpha tends to one.

**Current R126 author action.** The retained opposite tail-level behavior of the original lower and upper constants is preserved. The new stopping and support results state their own constants and interfaces.

**Evidence status.** Executed or proved components with explicit remaining scientific scope.

The main and supplement now state and prove finite rational-cut validity, residual correction over the declared box, the original-action regret enclosure and its finite-grid accuracy, and the direct-service projected gap, constrained optimizer bracket, separate regret/value-width gates and strict same-objective policy replacement. General convexity, interpolation and weak-duality ingredients are identified as inherited or classical.

**Remaining scientific requirement.** Matching an exponent at fixed alpha is not uniform matching as alpha approaches one.

**Source routes.** `new126/certified_support_main126.tex`, `new126/certified_support_proofs126.tex`, `new126/decision_stopping_main126.tex`, `new126/decision_stopping_proofs126.tex`.

### Minor 4. The readout coefficient bounds are acquisition restrictions, not characterizations of the best approximation; report sensitivity.

**Original request.** 4. The readout coefficient bounds are acquisition restrictions, not characterizations of the best approximation; report sensitivity.

**Inherited R125 amendment.** All inherited fitted actors, coefficient restrictions and acquisition bills remain frozen. The new finite value/gradient boxes are derived from valid original bounds and a derivative cap; they are not outcome-tuned representation restrictions. The source explicitly separates the exact information class from additional unencoded model formulas.

**Current R126 author action.** The coefficient box remains the historically prespecified bound 128, with the original signed/nonnegative restrictions. The independent audit verifies that box and the exact rounded readout; it does not choose a new bound after inspecting fit quality.

**Evidence status.** Executed or proved components with explicit remaining scientific scope.

The independently executed historical-data audit passed all 56 declared cases under F126. Of 24 VALUE solver calls configured for BVLS, four execute the distinct active-set path and 20 accept the feasible unconstrained solution through the shared initial least-squares path. There are also 24 independently assembled shifted MARGINAL LP solves and eight explicitly matched hinge copies, including four VALUE copies. All 28 VALUE rounded readouts reproduce the originals exactly; 39 of 56 readouts match overall, while all 17 different MARGINAL readouts and their objectives remain recorded. The maximum exact finite-box gap is below 2.34e-4 for VALUE sum squared loss and 2.84e-8 for the new MARGINAL residual. The whole external audit cost is 21.823981663 seconds, with zero economic queries and no deployed actor replaced.

**Remaining scientific requirement.** No new coefficient-box sensitivity campaign is claimed by the fixed-input audit.

**Source routes.** `FIT_AUDIT_PROTOCOL126.md`, `code126/independent_fit126.py`, `new126/independent_fit126.tex`, `FIT_AUDIT_REPORT126.md`, `new126/fit_results126.tex`, `new126/fit_development126.tex`, `results126/independent-fit-attempt01/EXECUTION126.json`, `results126/independent-fit-attempt01/SUMMARY126.json`, `results126/launch-fit-attempt01/PROCESS126.json`, `provenance126/FIT_NUMERICAL_PATH_READBACK126.json`.

### Minor 5. Distinguish validated residual span from Bellman action gap in terminology and tables.

**Original request.** 5. Distinguish validated residual span from Bellman action gap in terminology and tables.

**Inherited R125 amendment.** Residual span, original action contrast, optimizer bracket, original regret and absolute implemented-policy interval width are separate quantities throughout the new theorems and service protocol. An upper contrast must still include the unchanged future-controller allowance before an action is accepted.

**Current R126 author action.** The text and response distinguish validated residual span, finite-node value loss, finite-node action-marginal residual, Bellman action gap, original regret and implemented-policy value width. The certificate proofs specify which of these each bound controls.

**Evidence status.** Executed or proved components with explicit remaining scientific scope.

The independently executed historical-data audit passed all 56 declared cases under F126. Of 24 VALUE solver calls configured for BVLS, four execute the distinct active-set path and 20 accept the feasible unconstrained solution through the shared initial least-squares path. There are also 24 independently assembled shifted MARGINAL LP solves and eight explicitly matched hinge copies, including four VALUE copies. All 28 VALUE rounded readouts reproduce the originals exactly; 39 of 56 readouts match overall, while all 17 different MARGINAL readouts and their objectives

remain recorded. The maximum exact finite-box gap is below 2.34e-4 for VALUE sum squared loss and 2.84e-8 for the new MARGINAL residual. The whole external audit cost is 21.823981663 seconds, with zero economic queries and no deployed actor replaced.

The main and supplement now state and prove finite rational-cut validity, residual correction over the declared box, the original-action regret enclosure and its finite-grid accuracy, and the direct-service projected gap, constrained optimizer bracket, separate regret/value-width gates and strict same-objective policy replacement. General convexity, interpolation and weak-duality ingredients are identified as inherited or classical.

The frozen paid-support campaign executed all 74 pools, 888 contrasts and 5,920 LP calls. Exact independent reconstruction checked 33,331 cuts and 148 candidate regret records. It found 48 strictly sharper contrasts and six sharper regret bounds, but zero newly qualified policy gains and zero changed primary policies. No new economic query was made, and no finite solver optimality was certified. The six improved regret bounds concern capacity actions that remain unqualified, so they are not a delivered economic gain.

**Remaining scientific requirement.** One metric is not relabeled as another or used as an unsupported economic-loss proxy.

**Source routes.** `FIT_AUDIT_PROTOCOL126.md`, `code126/independent_fit126.py`, `new126/independent_fit126.tex`, `FIT_AUDIT_REPORT126.md`, `new126/fit_results126.tex`, `new126/fit_development126.tex`, `results126/independent-fit-attempt01/EXECUTION126.json`, `results126/independent-fit-attempt01/SUMMARY126.json`, `results126/launch-fit-attempt01/PROCESS126.json`, `new126/certified_support_main126.tex`, `new126/certified_support_proofs126.tex`, `new126/decision_stopping_main126.tex`, `new126/decision_stopping_proofs126.tex`, `results126/support01/SUMMARY126.json`, `results126/support-audit01/SUPPORT_VERIFICATION126.json`, `new126/support_results126.tex`, `provenance126/FIT_NUMERICAL_PATH_READBACK126.json`.

### Minor 6. Report rejection changes beside every span improvement.

**Original request.** 6. Report rejection changes beside every span improvement.

**Inherited R125 amendment.** The historical 36/36 span improvement is stated together with the rejection increase from 644 to 757. The new protocol reports action acceptance, failed precision and stopping outcomes with the costs that produced them.

**Current R126 author action.** Historical rejection changes accompany the span-improvement finding. The new complete census preserves first proposals, refusals, fallback, action decisions and work so action consequences can be read beside certificate changes.

**Evidence status.** Executed first-campaign evidence and proved components; the stated scientific limits remain open.

The frozen paid-support campaign executed all 74 pools, 888 contrasts and 5,920 LP calls. Exact independent reconstruction checked 33,331 cuts and 148 candidate regret records. It found 48 strictly sharper contrasts and six sharper regret bounds, but zero newly qualified policy gains and zero changed primary policies. No new economic query was made, and no finite solver optimality was certified. The six improved regret bounds concern capacity actions that remain unqualified, so they are not a delivered economic gain.

The current evidence reports all strict direct-versus-control gains, losses, exact ties and unresolved orderings beside certificate and work quantities. All 1,805 direct queries and 689 strict internal replacements remain recorded. A zero support-solver qualified-gain count is reported beside its 48 sharper contrast bounds and six sharper regret bounds.

**Remaining scientific requirement.** No favorable aggregate suppresses adverse method/state cells.

**Source routes.** `results126/support01/SUMMARY126.json`, `results126/support-audit01/SUPPORT_VERIFICATION126.json`, `new126/support_results126.tex`, `results126/report01/RESULTS126.md`, `results126/report01/SUMMARY126.json`, `results126/primary01/EXECUTION126.json`, `results126/independent01/SUMMARY126.json`.

### Minor 7. Label the acquisition bracket as an observed engineering range, not a confidence interval.

**Original request.** 7. Label the acquisition bracket as an observed engineering range, not a confidence interval.

**Inherited R125 amendment.** The minimum and maximum of the three R125 times are labeled descriptive shared-host timing brackets. The four-state census is deterministic and no confidence interval is constructed from these repeats.

**Current R126 author action.** Observed acquisition and timing brackets are identified as engineering ranges from recorded executions. New audit runtime is measured separately and belongs to research verification.

**Evidence status.** Executed first-campaign evidence and proved components; the stated scientific limits remain open.

The independently executed historical-data audit passed all 56 declared cases under F126. Of 24 VALUE solver calls configured for BVLS, four execute the distinct active-set path and 20 accept the feasible unconstrained solution through the shared initial least-squares path. There are also 24 independently assembled shifted MARGINAL LP solves and eight explicitly matched hinge copies, including four VALUE copies. All 28 VALUE rounded readouts reproduce the originals exactly; 39 of 56 readouts match overall, while all 17 different MARGINAL readouts and their objectives remain recorded. The maximum exact finite-box gap is below 2.34e-4 for VALUE sum squared loss and 2.84e-8 for the new MARGINAL residual. The whole external audit cost is 21.823981663 seconds, with zero economic queries and no deployed actor replaced.

The actual three-repetition timing ranges and preserved acquisition brackets are observed engineering ranges. The separate fitting audit cost of 21.823981663 seconds is research verification on the shared host. No coverage probability is assigned to these clocks or brackets.

**Remaining scientific requirement.** These ranges are not confidence intervals for future hardware or populations.

**Source routes.** FIT_AUDIT_PROTOCOL126.md, code126/independent_fit126.py, new126/independent_fit126.tex, FIT_AUDIT_REPORT126.md, new126/fit_results126.tex, new126/fit_development126.tex, results126/independent-fit-attempt01/EXECUTION126.json, results126/independent-fit-attempt01/SUMMARY126.json, results126/launch-fit-attempt01/PROCESS126.json, results126/report01/RESULTS126.md, results126/report01/SUMMARY126.json, results126/primary01/EXECUTION126.json, results126/independent01/SUMMARY126.json, provenance126/FIT_NUMERICAL_PATH_READBACK126.json.

### Minor 8. State visibly that there are zero adoption regions over quadratic least squares.

**Original request.** 8. State visibly that there are zero adoption regions over quadratic least squares.

**Inherited R125 amendment.** The zero R97 neural adoption count over quadratic least squares is retained explicitly with its original 1,800-cell sensitivity scope. R125 defines its own full-menu sufficient and optimistic comparisons on a matching first-state workload and does not transplant older regions into the new experiment. The separate post-execution Proposition policyidentity125 establishes the stronger conditional exclusion available when actual complete-policy equality and strictly lower conventional complete costs are known.

**Current R126 author action.** The R97 finding of zero certified adoption regions over quadratic least squares is retained. R125 negative identity/cost results also keep their original episode. The new experiment is reported under its own source and selection rule.

**Evidence status.** Executed first-campaign evidence and proved components; the stated scientific limits remain open.

The historical zero-adoption finding is preserved with its original scope. The new first campaign also has zero sufficient neural adoption regions across all 96 conditions, with exact same-policy conventional cost blockers in all 96. All optimistic outer regions remain nonempty, which is reported instead of being discarded.

The C episode supplies 192 sufficient neural adoption regions, all empty, with 192 exact same-policy conventional cost blockers; all 192 optimistic outer regions remain nonempty. Exact reuse improves common certificate computation without establishing a neural-specific representation, information or policy advantage.

**Remaining scientific requirement.** There is no inherited or assumed neural adoption claim over the full conventional menu.

**Source routes.** results126/report01/RESULTS126.md, results126/report01/SUMMARY126.json, results126/primary01/EXECUTION126.json, results126/independent01/SUMMARY126.json, SOURCE_FREEZE126C.json, BOUNDARIES126C.json, new126/cached_envelopes_main126c.tex, new126/cached_envelopes_proofs126c.tex, results126c/verification01/EXECUTION126.json, results126c/report01/SUMMARY126C.json, results126c/report01/RESULTS126C.md.

### Minor 9. Compare the certification contribution carefully with existing verified approximation and HJB-residual literatures.

**Original request.** 9. Compare the certification contribution carefully with existing verified approximation and HJB-residual literatures.

**Inherited R125 amendment.** The complete-support theorem is placed in relation to smooth convex interpolation and the inherited verified Bellman/HJB literature. Its contribution is the exact finite retention of the original global support payload and its action-contrast interpretation, with a constructive proof and a rational residual correction. The stochastic-control and nonlinear-recursive bridges retain their own assumptions.

**Current R126 author action.** The paper identifies the inherited verified-approximation, HJB-residual and information-complexity ingredients separately from its new original-action calculation. Exact finite-box audit bounds are credited to convexity and weak duality, with complete proofs supplied.

**Evidence status.** Executed or proved components with explicit remaining scientific scope.

The main and supplement now state and prove finite rational-cut validity, residual correction over the declared box, the original-action regret enclosure and its finite-grid accuracy, and the direct-service projected gap, constrained optimizer bracket, separate regret/value-width gates and strict same-objective policy replacement. General convexity, interpolation and weak-duality ingredients are identified as inherited or classical.

The frozen paid-support campaign executed all 74 pools, 888 contrasts and 5,920 LP calls. Exact independent reconstruction checked 33,331 cuts and 148 candidate regret records. It found 48 strictly sharper contrasts and six sharper regret bounds, but zero newly qualified policy gains and zero changed primary policies. No new economic query was made, and no finite solver optimality was certified. The six improved regret bounds concern capacity actions that remain unqualified, so they are not a delivered economic gain.

**Remaining scientific requirement.** A shared certificate mechanism or known convex construction is not presented as neural-specific novelty.

**Source routes.** `new126/certified_support_main126.tex`, `new126/certified_support_proofs126.tex`, `new126/decision_stopping_main126.tex`, `new126/decision_stopping_proofs126.tex`, `results126/support01/SUMMARY126.json`, `results126/support-audit01/SUPPORT_VERIFICATION126.json`, `new126/support_results126.tex`.

### Minor 10. The title should not carry more neural-specific content than the theorem and evidence identify.

**Original request.** 10. The title should not carry more neural-specific content than the theorem and evidence identify.

**Inherited R125 amendment.** The title retains Neural Bellman Operators, as the original learned recursive-operator research identity requires. The text states which theorems are shared with conventional representations and which comparisons would be needed to establish neural-specific value.

**Current R126 author action.** The original Neural Bellman Operators title and economic research identity are retained. The text describes which objects are neural, which guarantees are shared and which comparative benefits the data actually identify.

**Evidence status.** Executed first-campaign evidence and proved components; the stated scientific limits remain open.

The current title remains Neural Bellman Operators. The actual catalogue includes neural continuation critics and actors, matched conventional representations and original economic losses. The source-bound first campaign establishes no neural-specific complete-cost advantage, and the title is not used to imply one.

**Remaining scientific requirement.** The title is not treated as evidence of neural-specific dominance; unsupported superiority statements remain excluded.

**Source routes.** `results126/report01/RESULTS126.md`, `results126/report01/SUMMARY126.json`, `results126/primary01/EXECUTION126.json`, `results126/independent01/SUMMARY126.json`.